

AI-Powered Visual Book Generator



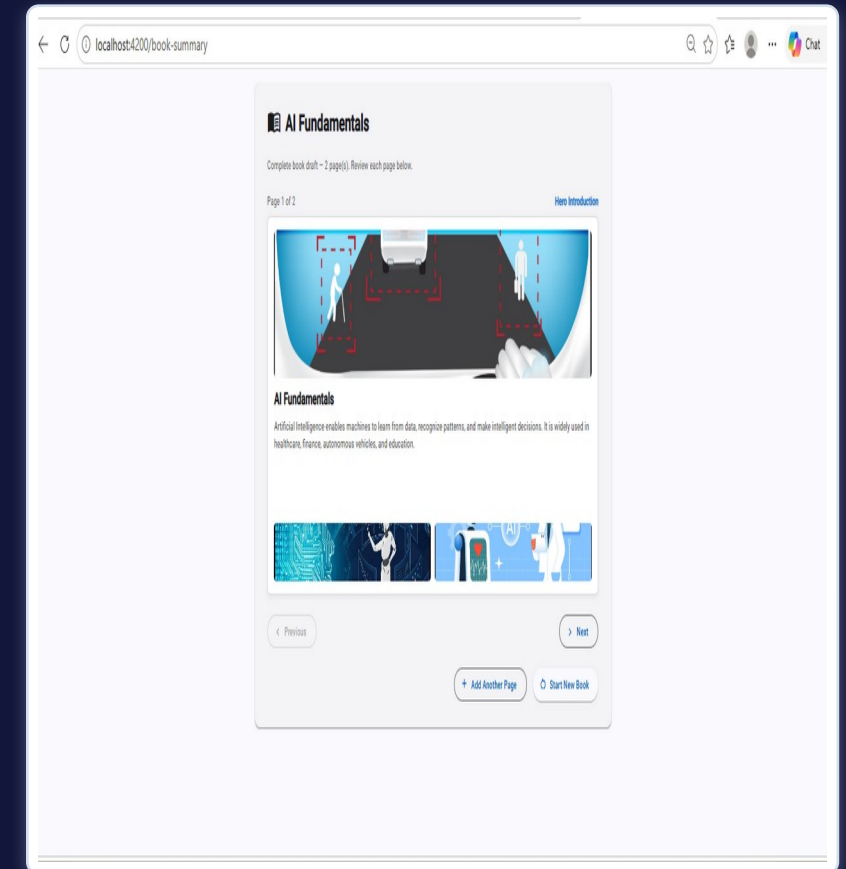
Aman Agrawal

~11+ years in software engineering · M.Tech in AI & ML, IIT Jodhpur

aman.agrawal@telus.com

linkedin.com/in/aman-agrawal007

github.com/Aman007Agg/ai-powered-visual-book-generator



Problem Statement

Marketing and content teams create content-heavy documents that take significant manual effort to turn into visually appealing layouts.

Designers spend considerable time arranging text and images and iterate many times before producing a polished document.

The objective is to automate the first-draft generation using Large Language Models — cutting manual effort while keeping layout recommendations context-aware.

Today's cost

- Heavy manual effort
- Multiple design iterations
- Significant designer time
- Documents up to ~500 pages

Project Objectives

1

Browser-based app

Runs entirely in the browser — no install.

2

Accept book context

Optional title and objective guide the AI.

3

Accept page content

Page-by-page text input.

4

Accept images

Upload one or more images per page.

5

Two layout options

AI generates two intelligent options per page.

6

User selects

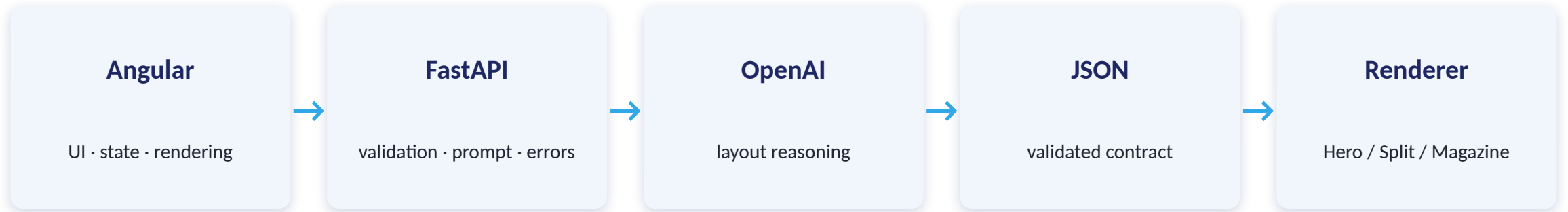
The user picks the preferred layout.

7

Complete draft

Continue page by page to a full book draft.

Architecture



Why the LLM never generates HTML or CSS

The model decides layout SEMANTICS as JSON; Angular owns PRESENTATION. That separation means: no markup from the model = no XSS/injection surface · deterministic, testable rendering · restyling costs no AI call · the same JSON could later drive PDF/print.

Technology Stack

Angular

- Modern SPA
- Component architecture
- Strong TypeScript support
- Signals + zoneless

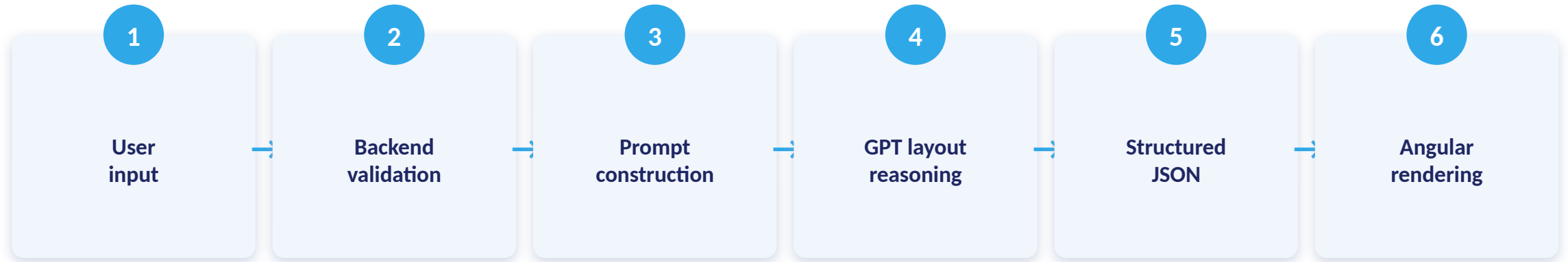
FastAPI

- High performance, async
- Pydantic validation
- Clean REST APIs
- Typed request/response

OpenAI

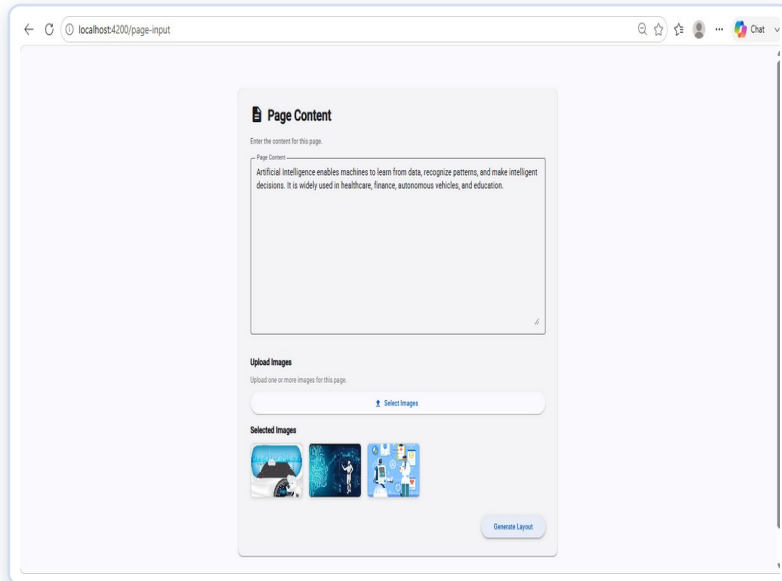
- Strong reasoning
- Structured JSON outputs
- Vision-capable (roadmap)
- Prompt engineering

AI Workflow

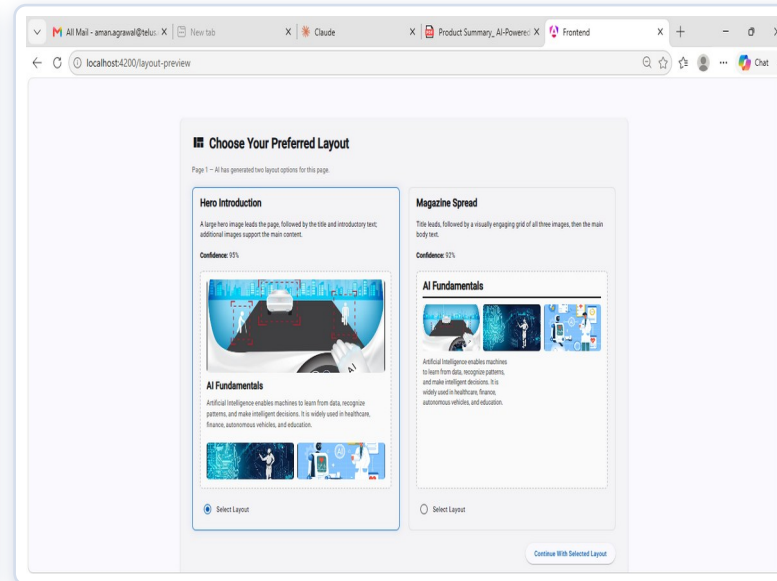


Separation of concerns: the AI focuses only on reasoning and layout decisions; Angular is solely responsible for presentation. The result is safe, predictable, and easy to test.

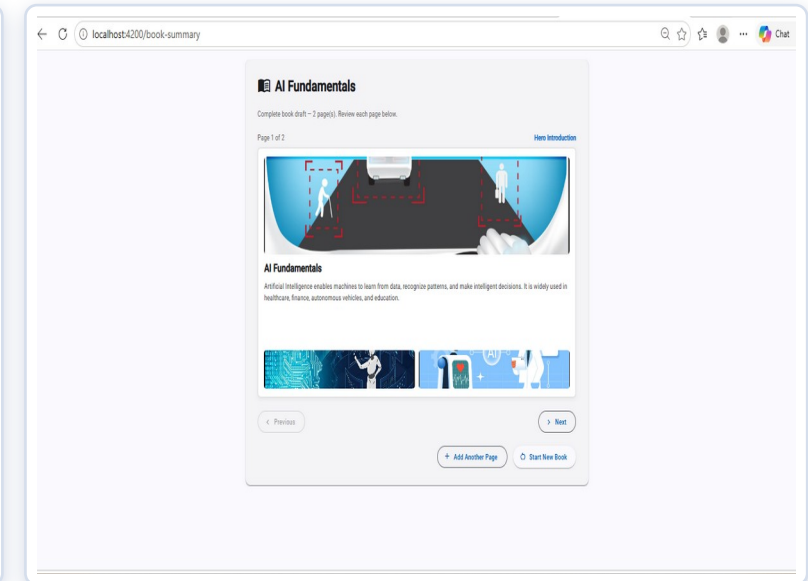
Live Demo



Page content + images



Two AI layout options



Complete book draft

Demo flow: New Book → book details → add page → upload image → generate layout → two options → select → continue → final book.

Engineering Decisions

Page-level processing

Keeps each request within LLM context limits (~500-page books).

Structured JSON output

Validate-able, safe, deterministic — no free-form parsing.

FastAPI

Async, high-performance, first-class Pydantic validation.

Angular

Typed, component-driven SPA with Signals for clean state.

No database (MVP)

PRD non-goal; state is in-memory for a client-side prototype.

No drag-and-drop (MVP)

PRD non-goal; focus on AI layout intelligence first.

Future Enhancements

1

Natural-language editing

2

Themes

3

Persistent storage

4

Multi-user collaboration

5

PDF export

6

Style consistency

7

Batch processing

8

Template library

Thank You

Thank you for your time. I appreciate the opportunity to work on this assignment and I'm happy to discuss any architectural decision or implementation detail.



Aman Agrawal

aman.agrawal@telus.com

[linkedin.com/in/aman-agrawal007](https://www.linkedin.com/in/aman-agrawal007) · github.com/Aman007Agg/ai-powered-visual-book-generator